Email DLP – Human-in-the-Loop Machine Learning Architecture

This document describes the optimal architecture and step-by-step workflow for building a continuous learning, human-in-the-loop Machine Learning system for Email Data Loss Prevention (DLP) using Netskope logs.

Recommended Approach

The optimal approach is a Python-based Machine Learning pipeline with Splunk acting as the log ingestion, visualization, and feedback interface. This provides flexibility, model control, and scalability while avoiding tight coupling with Splunk MLTK limitations.

High-Level Workflow

- Email events are captured by Netskope and ingested into Splunk.
- Relevant email metadata is extracted and transformed into ML features.
- An ML model clusters and classifies incidents into risk bins.
- Analysts review classifications and provide feedback.
- Feedback is used to retrain the model.
- Known patterns are auto-classified; unknown patterns are flagged for review.

Step-by-Step Detailed Architecture

Step 1: Log Ingestion
Netskope email DLP logs are ingested into Splunk with standardized fields (sender, recipient, domain types, CC, policy, user type, timestamp).

Step 2: Feature Engineering
Logs are exported to a Python pipeline where categorical and numerical features are encoded for ML consumption.

Step 3: Initial Pattern Learning
Unsupervised or semi-supervised models group incidents into behavioral clusters representing risk patterns.

Step 4: Analyst Feedback Loop
Analysts validate clusters, assign severity, and mark incidents as valid or false positives.

Step 5: Supervised Re-training
Analyst feedback is used as labeled data to improve model accuracy and confidence.

Step 6: Auto-Classification
The model automatically classifies known patterns and closes low-risk incidents.

Step 7: Novel Pattern Detection
Low-confidence or unseen patterns are flagged and routed for human analysis, ensuring

adaptability to new risks.

Final Outcome

This architecture ensures reduced analyst workload, adaptive risk detection, and continuous improvement of DLP decision-making while maintaining human oversight for new and high-risk patterns.